# Bo Li

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## Research Interests

AI-enabled chemical discovery; quantum chemistry, statistical mechanics, and molecular simulation; chemical knowledge and coordination equilibria for critical-mineral recovery; generative molecular design and reaction networks for reversible $CO_2$ capture; agentic screening and collective transport in disordered ionic materials; molecular mechanisms in catalytic conversion; interpretable machine learning and adaptive simulation.

## Academic Appointments

**Research Assistant Professor**, Chemical and Biomolecular Engineering, Vanderbilt University — Jan. 2023 – present

**Postdoctoral Research Associate**, Theoretical Division, Los Alamos National Laboratory — Jan. 2020 – Jan. 2023

**Postdoctoral Scholar**, Department of Chemistry, University of California, Riverside — Nov. 2016 – Jan. 2020

## Education

**Ph.D., Theoretical Chemistry** — 2015
Technical University of Munich, Germany. Advisor: Prof. Dr. Notker Rösch.
*Computational Modeling of Chemical Systems in Solution: A Microscopic Approach for Efficient Density Functional Calculations with ParaGauss*

**M.Eng., Material Physics and Chemistry**, Graduate University of Chinese Academy of Sciences, China — 2011

**B.Sc., Geophysics**, Peking University, China — 2008

## Research Experience

**Vanderbilt University** — Jan. 2023 – present

- Lead research connecting electronic structure, intermolecular interactions, statistical mechanics, and transport in ionic liquids, molten salts, coordination compounds, and disordered materials.
- Develop generative molecular design for $CO_2$-reactive carbanions and interpretable, data-efficient ML models; integrate ab initio molecular dynamics, polarizable and machine-learned potentials, and enhanced sampling.
- Develop the REEL-QUEST chemical knowledge base and prototype agentic reaction workflows for chemically grounded discovery.
- Collaborate with experimental researchers to interpret spectroscopy, structural characterization, adsorption, and transport measurements.

**Los Alamos National Laboratory** — Jan. 2020 – Jan. 2023

- Developed density-functional tight-binding parameters for f-block chlorides and integrated DFTB with machine learning to enable larger molecular dynamics simulations.
- Modeled molten-salt and f-element systems for nuclear energy applications; investigated molten lanthanide chloride structures with ab initio molecular dynamics.

**University of California, Riverside** — Nov. 2016 – Jan. 2020

- Investigated uranyl binding with amidoxime ligands using molecular dynamics and enhanced free-energy sampling for seawater uranium extraction.
- Developed polarizable models of molten salts; applied DFT, MD, and QM/MM to $CO_2$ capture and electrocatalysis; contributed to DOE-NEUP proposals and technical reports.

## Selected Publications

51 publications and book chapters. Full list on Google Scholar.

1. **Li, B.**; Jiang, D.-e. "Discovering $CO_2$-Reactive Carbanions via Property-Guided Generative AI." *J. Chem. Inf. Model.* **2026**, 66, 6983–6991.
2. **Li, B.**; Fu, Y.; Yang, Z.; Dai, S.; Jiang, D.-e. "Intermolecular Proton Transfer Enabled Reactive $CO_2$ Capture by the Malononitrile Anion." *J. Phys. Chem. B* **2024**, 128, 10207–10213.

3. Qiu, L.; **Li, B.**; Hu, J.; Ganesan, A.; Pramanik, S.; Damron, J. T.; Li, E.; Jiang, D.-e.; Mahurin, S. M.; Popovs, I.; Steren, C. A.; Fan, J.; Yang, Z.; Dai, S. "Cascade $CO_2$ Insertion in Carbanion Ionic Liquids Driven by Structure Rearrangement." *J. Am. Chem. Soc.* **2024**, 146, 29588–29598.
4. Zhu, C.; **Li, B.**; Li, C.; Lu, L.; Li, H.; Yuan, X.; Kang, X.; Jiang, D.-e.; Zhu, M. "Arylation of Gold Nanoclusters and Insights into Structure-related $CO_2$ Reduction Reaction Performances." *Chem. Sci.* **2025**, 16, 10273–10281.
5. Pramanik, S.; **Li, B.**; Driscoll, D. M.; Johnson, K. R.; Evans, B. R.; Damron, J. T.; Ivanov, A. S.; Jiang, D.-e.; Einkauf, J.; Popovs, I.; Jansone-Popova, S. "Tetradentate Ligand's Chameleon-Like Behavior Offers Recognition of Specific Lanthanides." *J. Am. Chem. Soc.* **2024**, 146, 25669–25679.
6. Lei, M.; **Li, B.**; Liu, H.; Jiang, D.-e. "Dynamic Monkey Bar Mechanism of Superionic Li-ion Transport in $LiTaCl_6$." *Angew. Chem. Int. Ed.* **2024**, 63, e202315628.

## Teaching and Mentorship

- Developed and deployed computational infrastructure for a molecular simulation course for senior undergraduate and graduate students.
- Organized high-performance computing and molecular simulation workshops for chemistry students.
- Mentored graduate, undergraduate, and high-school researchers in computational chemistry, molecular simulation, and scientific programming.
- Teaching interests: general and physical chemistry; quantum chemistry, statistical mechanics, molecular thermodynamics and simulation; materials chemistry and scientific AI.

## Funding and Proposal Experience

Contributed computational strategy, technical sections, simulation plans, and HPC justification to multi-investigator proposals supported by DOE Basic Energy Sciences and DOE Nuclear Energy, including NEUP. Experience developing research concepts, revising proposals, and responding to reviewer feedback.

## Presentations

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| Feb. 2024 | Poster, Gordon Research Conference on Atomically Precise Nanochemistry, Galveston, TX. |
| Oct. 2023 | Invited talk, Materials Science & Technology Conference, Columbus, OH. |
| Feb. 2023 | Poster, Theory Frontiers in Actinide Science: Chemistry & Materials, Santa Fe, NM. |
| Jul. 2022 | Poster, 19th Chinese-American Kavli Frontiers of Science Symposium, Irvine, CA. |
| Feb. 2020 | Oral, International Workshop on Theory Frontiers in Actinide Sciences, Santa Fe, NM. |
| Aug. 2019 | Oral, ACS Fall National Meeting, San Diego, CA. |
| Jun. 2019 | Oral, ANS Annual Meeting, Minneapolis, MN. |
| May 2015 | Poster, Bunsentagung, Bochum, Germany. |
| Mar. 2014 | Oral, TUM-A*STAR Workshop on Computational Catalysis, Garching, Germany. |

## Professional Service

Journal reviewer: *Journal of Materials Chemistry C*, *The Journal of Physical Chemistry*, *Dalton Transactions*, *Nanoscale*, *Small Methods*, *ACS Applied Energy Materials*, *Applied Surface Science*, *Journal of Molecular Liquids*, *Advanced Science*, *Chemical Engineering Journal*, *Journal of Solution Chemistry*, and *Microporous and Mesoporous Materials*.

Reviewer of high-performance computing allocation proposals at Los Alamos National Laboratory. Member of the American Chemical Society, American Nuclear Society, and American Ceramic Society.

## Honors and Awards

TUM International Graduate School of Science and Engineering Scholarship (2012); NIMTE Outstanding Student Scholarship (2010); CAS Graduate Student National Scholarship (2008).

## Computational Expertise

**Electronic structure and simulation:** CP2K, VASP, Quantum ESPRESSO, DFTB+, ORCA, LAMMPS, GROMACS, AMBER.

**Machine learning:** PyTorch, TensorFlow, scikit-learn, Chemprop, Gaussian process regression, neural-network potentials, symbolic regression, diffusion models, and variational autoencoders.

**Programming and HPC:** Python (NumPy, SciPy, Pandas, ASE), Fortran, C/C++, Bash, MPI, CUDA, SLURM/PBS, Linux.